

Cold Calculations - How Conflict Characteristics Shape External Support in Armed Conflicts

Celine Giese and Thiago R. Oliveira

Contents

Table of Figures

Data availability statement

Abstract

Introduction

Literature Review

The study of armed conflicts in Criminology	
External support to armed conflicts	

Research Design

Methodology

Data and operationalisation	
Data analysis	

Findings

Overall trends in conflict characteristics	
The provision of external support	

Discussion

Conclusion

Appendices

References

Table of Figures

Type	Title	Page
Tables		
Table 1	Variables used in the analysis	13
Table 2	Regression Results – External Support Determinants	20
Table 3	Statistical Significance in the RE and FE Logistic Regression Models	21
Table 4	RE and FE Logistic and PPML Regression Results	24
Figures		
Figure 1	Directed Acyclic Graph on Conflict Characteristics and External Support	10
Figure 2	World armed conflicts by intensity level	18
Figure 3	Mean and Median Conflict Duration by Region	18
Figure 4	Direct and Indirect Support, 1975–2017	19
Appendices		
Appendix 1	State-based Conflicts by Region, 1975–2017	26
Appendix 2	State-based Conflicts by Type of Conflict, 1975–2017	26
Appendix 3	Conflict Type by Region	27
Appendix 4	Evolution of State-based Conflicts by Incompatibility, 1975–2017	27
Appendix 5	Evolution of External Support: Bilateral vs Coalition	28
Appendix 6	Statistical Significance in the RE Linear Regression models	29

Data availability statement

All analyses were conducted in R (R Core Team (2025)), with the analytic code and input data made publicly available on GitHub to ensure transparency and facilitate replication.

Abstract

Armed conflicts shape not only military but also social priorities, often diverting resources from social welfare to defence (Sivard (1996)). Their consequences can be devastating and long-lasting, affecting economic development, social structures, and state stability. As contemporary conflicts continue to unfold globally, understanding how and why states choose to intervene - particularly through external support (ES) - is increasingly relevant. Such interventions raise critical questions about international responsibility and complicity, especially when support facilitates war crimes. Despite this, armed conflict has received limited attention within Criminology. This study seeks to address that gap by investigating whether and how conflict characteristics influence the provision and form of ES. Drawing on 472 dyads in 212 conflicts between 1975 and 2017, sourced from the Uppsala Conflict Data Program (UCDP), the analysis is guided by a Directed Acyclic Graph (DAG) that maps assumed causal pathways. The study employs within-between logistic regression models to test hypotheses on the influence of conflict characteristics - such as intensity, duration, and type - on the provision of ES, including whether support is given bilaterally or through coalitions. Based in criminological research on violence (Collins (2008)) this suggests a “cold” approach to the provision of external support, where external supporters consider the conflict dynamics before providing support. This research contributes to a more comprehensive understanding of the interplay between external interventions and conflict dynamics, bridging critical gaps in criminology and offering a starting point for criminological discussion around the liability and responsibility of external supporters.

Introduction

Despite a reduction in conflicts and deaths, the past decade remains the most violent on record (Davies et al. (2024)). War and armed conflicts (AC) profoundly impact parties involved, with ordinary civilians disproportionately suffering due to fatalities, displacement, and long-term public health crises from destroyed infrastructure (Hoeffler and Reynal-Querol (2003)). War-torn countries face reduced growth, higher poverty rates, and weakened institutions, with recovery dependent on robust policy reforms (Hoeffler and Reynal-Querol (2003)). High-intensity violence during conflicts lowers economic growth, life expectancy, and educational attainment, with these effects often persisting long-term (Le et al. (2022)). Wars and ACs further generate significant spillover effects, with severity influenced by geographic proximity and economic ties to the conflict zone. Neighbouring countries often face adverse economic consequences, while distant nations experience milder effects due to limited trade links and geographic distance (Federle et al. (2024)). Regions with stronger economic ties to the conflict zone experience amplified negative impacts, including disrupted stability and heightened uncertainty (Fang et al. (2020); Le et al. (2022)). While inflation and output remain stable in distant countries, the ripple effects of war challenge interconnected global economies, highlighting the complex dynamics between conflict and economic stability (Federle et al. (2024)). The enduring economic and social costs of war underscore its profoundly criminogenic nature.

Today, although conflicts are largely fought by developing nations, they rely heavily on advanced military equipment produced by industrial countries, reinforcing military over social investments (Sivard (1996)). The prevalence and complexity of external support (ES) in AC have grown significantly over the past decades. From 1975 to 2017, 80% of intrastate conflicts involved at least one instance of ES, with state actors being the predominant supporters (Meier et al. (2023)). Recent years have seen systemic shifts in ES, characterised by the rise of multilateral coalitions and collaborative interventions. Nearly one-third of all support instances now involve such coalitions, emphasising burden-sharing and the legitimacy of support (Meier et al. (2023)). However, recent developments, such as the United States cutting off significant amounts of its financial and military support for Ukraine, and humanitarian aid to Syria (Sandefur and Kenny (2025); UN (2025)), cast doubt on the future of military alliances and joint defence initiatives like the North Atlantic Treaty Organisation (NATO), as well as the accountability and durability of assistance. With many geopolitical tensions and conflicts ongoing, discussions around AC and ES are crucial to the political environment and international relations, raising questions around the responsibility and accountability of external supporters, especially when military actions turn into war crimes.

While an extremely timely topic, Criminology has historically neglected AC (Hagan (2015); Jamieson (2003); Ruggiero (2015); McGarry and Walklate (2015)), let alone ES. But even in research on ES, the content of interventions remains underexplored, despite its impact on conflict trajectories (see for example Cunningham (2016); Karlén (2017); Aydin and Regan (2012)). This research adopts an interdisciplinary approach, integrating Criminology, Peace and Conflict Studies, and Economics to examine whether conflict characteristics affect the provision of ES and identify trends and conditions under which ES is provided. While Economics tend to focus on the economic costs of conflicts, using quantitative analysis to analyse the economic effects and spillover of conflicts (see for example (Federle et al. (2024); Fang et al. (2020); Sivard (1996))), Peace and Conflict studies predominantly use qualitative approaches, focusing on the prevention, resolution, and transformation of conflict (i.e. *INSERT REFERENCE*). Critical Criminology, which has started to engage with AC primarily does so from a theoretical and/or qualitative approach, predominantly focusing on the criminogenic nature of war as a whole (i.e. Ruggiero (2005)) and genocide (i.e. Rafter (2016)). Through drawing on literature from Economics, as well as Peace and Conflict studies, this study aims to combine the quantitative approach of Economics with the focus areas of Peace and Conflict studies and criminological theories to offer broader insights into the areas of ES and AC. By distinguishing between support types, it further aims to provide a more robust understanding of the role of external supporters and their motives, essential for further discussions around liability and responsibility. In doing so, it bridges critical gaps in Criminology, contributing to a more comprehensive understanding of the interplay between external interventions and conflict dynamics.

This study addresses three research questions: (1) Do conflict characteristics influence the provision of ES? (2) Do they affect the form of support provided? (3) Does coalition-based support influence what type

of support is provided? The data comes from two Uppsala Conflict Data Program (UCDP) datasets: the Dyadic Dataset (Version 18.1), covering conflict characteristics between 1946 and 2018 (Harbom et al. (2008); Pettersson and Eck (2018)), and the External Support Dataset (Version 18.1), which captures ten forms of ES between 1975 and 2017 (Meier et al. (2023)). The high-quality data produced by UCDP is methodically gathered, covers mass violence globally, and includes long-term series that are updated once a year. It is the oldest civil war data collection and allows for comparisons between conflicts and nations. Its definition of AC has evolved into the accepted worldwide norm for the methodical definition and analysis of conflicts. Together, the two datasets provide a robust foundation for quantitatively analysing the relationship between conflict characteristics and ES. To answer the research questions, the study uses within-between logistic regression models, allowing for the examination of both within- and between-dyad variation. These methods are well suited to the dyadic nature of AC data and support the identification of strategic patterns in support provision—insights essential for discussions on the liability and responsibility of external actors. Therefore, this study contributes to a more nuanced understanding of the interplay between external interventions and conflict dynamics, addressing critical gaps in criminological literature on accountability in AC. The study is structured as follows: First, existing literature on AC and ES is reviewed, highlighting gaps in criminological scholarship and introducing the conceptual lens of ‘hot’ versus ‘cold’ mass violence. The research design and theoretical framework are then described, followed by the methodology, data, and operationalisation. Following the presentation of the findings, it finishes with the discussion and concluding reflections.

Literature Review

The landscape of AC has shifted in recent decades, with a rise in internationalised intrastate conflicts where external states support non-state actors opposing governments (Davies et al. (2024)). This study uses the UCDP’s definition of AC, which has become a norm for the methodical definition and analysis of conflicts. AC herein is defined as ‘a contested incompatibility that concerns government and/or territory where the use of armed force between two parties, of which at least one is the government of a state, results in at least 25 battle-related deaths in a calendar year’ (Themnér (2018), p.2). The primary parties in such conflicts include governments, which control the state capital, and opposition organisations, which are formally organised groups using armed force to influence the outcome of the incompatibility.

Although developing nations fight most of today’s conflicts, they are often fought with weapons from industrialised countries, reflecting 20th-century technological advancements (Sivard (1996)). As geopolitical tensions rise, the role of ES in these conflicts has become a central topic of discussion (Federle et al. (2024)). ES, defined as militarily relevant assistance provided by outside parties to active conflicts, is distinct from peacetime activities like arms trade or security reforms (Meier et al. (2023); Meier (2022)). This support aims to enhance the recipient’s military capabilities and is provided with the clear intent of facilitating military victory over the opposing side (Cunningham (2010); Gregory (2000)). It includes materiel, knowledge, or services directly aiding warfare but excludes unintentional support like sanctions that weakens a conflict party but is not intended to support the other side. Additionally, diplomatic efforts, humanitarian aid, and mediation offers do not qualify as ES since they do not directly contribute to the military outcome (Meier (2022)). The external nature of support is determined by whether the provider is not a primary warring party in the conflict, even if present in the same territory.

The interplay between conflict characteristics and ES is complex, with interventions - whether direct military actions or indirect aid - often prolonging hostilities and complicating peace efforts (Sawyer et al. (2015); Karlén (2023); Regan (2002)). Understanding why and how states intervene and the consequences of different types of support is vital to analysing external involvement. This research seeks to explore these themes, examining how conflict characteristics shape external interventions.

The study of armed conflicts in Criminology

Following the so-called ‘century of peace’ (1815–1914), war has continued across regions, yet social sciences have failed to reflect its prevalence or significance. States, through their ‘legitimate monopoly of violence’

(Weber (1919) in McGarry and Walklate (2019)), justify violence via bureaucratic rationality, framing war as a noble endeavour to solidify authority and autonomy (Malešević (2010) in Walklate and McGarry (2019)). Historically, war was driven by imperial capitalist expansion, which fuels conflicts through profit-driven globalisation (Ruggiero (2023); Bonger (1916) in Walklate and McGarry (2019); Klein (2011)). Even today, economic imperatives often guide responses to war, reducing it to a developmental obstacle. The global capitalist system blurs lines between legitimate but harmful actions, exacerbating violence (Ruggiero (2005)). Critical criminologists therefore view war as a form of state crime legitimised by dominant powers who shape international norms to justify past and future conflicts (Kramer and Michalowski (2005); Bonanate (2011) in Ruggiero (2023)). Despite millions of legitimate killings in war, criminology has been reluctant to challenge the legitimacy of such violence or address its broader systemic causes (Ruggiero (2005)). Wartime atrocities are reframed as patriotic acts, aligning with state narratives through techniques of neutralisation (Sykes and Matza (1957); Ruggiero (2005)). These dynamics underscore the importance of rethinking state authority, as legality and illegality are often strategically maneuvered in the pursuit of power (Heyman (1999) in Ruggiero (2005)). Criminologists, long defining crime not only over illegality but over harm (Hillyard and Tombs (2007)), have a unique opportunity to address its multifaceted impacts and advocate for systemic reforms (Jamieson (1998) in McGarry et al. (2016); Ruggiero (2005)). Due to its central role to conflicts, examining ES is essential to understanding underlying dynamics of AC. It further raises critical questions about complicity and liability for war crimes, and whether ES escalates violence or fosters peace.

To examine the dynamics of ES, offering insights into the motives and strategies of external intervention, this study analyses ES through the lens of ‘hot’ and ‘cold’ conflicts, a principle first developed by Collins (Collins (2008)). According to Collins (2008) in (Rafter (2016), p.98), whether fighting actually breaks out in violent confrontations between individuals or groups depends on ‘a series of conditions or turning-points that shape the tension and fear in particular directions, reorganising the emotions as an interactional process involving everyone present: the antagonists, audience, and even ostensibly disengaged bystanders’. These turning points operate as triggers for later (but still contingent) behaviours and signal a break with the feelings of the past. Criminological studies on mass violence (MV) have looked at the situational dynamics of such acts, distinguishing between ‘hot’ and ‘cold’ acts of MV (Rafter (2016)), reflecting their underlying emotional dynamics. At the core of ‘hot’ MV is the notion of ‘forward panic’ - a form of violence that emerges when prolonged fear and tension build up, then suddenly discharge in a wave of uncontrolled aggression. This pattern is marked by emotional entrainment: participants get caught in a rhythm of escalating violence that feels unstoppable in the moment (Collins (2008)). Emotions such as rage, fear, elation, and a sense of moral detachment combine to create a state of high physiological and social arousal. In contrast, ‘cold’ violence is premeditated, detached, and bureaucratised. It reflects deliberate planning and instrumental logic, often implemented through structures of command or strategy. While it lacks the overt emotional frenzy of forward panics, it is no less violent - merely more strategic, procedural, and distanced from immediate confrontation. These categories are not mutually exclusive: conflicts may shift between ‘hot’ and ‘cold’ phases or contain elements of both. Importantly, Collins (2008) theory treats emotional dynamics not as irrational noise but as socially organised processes with identifiable triggers and consequences. Applying this lens to the provision of ES, this study hypothesises that external supporters - distant from the emotional immediacy of the battlefield - act more in line with ‘cold’ violence logics. Support decisions are hypothesised to reflect calculated responses to conflict characteristics such as intensity, duration, type, or geopolitical timing (e.g., post-9/11). By testing whether certain conflict characteristics are correlated with the overall provision and type of ES, the study examines whether ES is best understood as a strategic, interest-driven act rather than an emotionally reactive one.

External support to armed conflicts

Despite only a slight rise in total conflicts over recent decades, the number of actors providing ES has grown (Meier et al. (2023)). While troop support has declined, the total number of external interventions continues to grow, reflecting persistent interest in shaping conflict trajectories through both direct and indirect means (Davies et al. (2024); Chang and Sellak (2022)). By 2017, 77% of active conflict-dyads featured state support for governments alone, driven in part by the post-9/11 counter terrorism focus, which stigmatised support for non-state actors (Meier et al. (2023)). Multilateral counter-extremism efforts, often

led by UN Security Council members, have reinforced this trend (Meier et al. (2023)). Research on ES in ACs highlights its significant impact on conflict dynamics and outcomes. The provision of ES shapes the onset (Cunningham (2016); Regan and Meachum (2014)), duration (Aydin and Regan (2012); Anderson (2019)), outcomes (Sawyer et al. (2015)), and recurrence (Karlén (2017)) of ACs. Interventions, whether direct (troops) or indirect (weapons, logistics, training), frequently prolong conflicts by altering the cost-benefit calculations of belligerents (Fearon (2004) in Testerman (2015); Lacina (2006)) and although external aid can alter power dynamics, it frequently intensifies conflicts, complicates negotiations, and reduces the likelihood of peaceful resolutions (Olson Lounsbery (2016); Zartman (1992) in Saideman (2002); Salehyan et al. (2011); Petrova (2019)). Although ES can improve the odds of military victory (Balch-Lindsay et al. (2008)), it often exacerbates humanitarian crises, with rebels exploiting moral hazards to attract further intervention (Rauchhaus (2009); Karlén (2017)). Overall, external actors, including state and non-state supporters, play pivotal roles by shaping conflict processes through diverse forms of assistance, such as military training and direct intervention (Gleditsch (2007); Salehyan (2010); Toukan (2019)).

The type and target of ES often align with intervening parties' strategic interests. While training and expertise remain the most frequent forms of support, material aid dominates assistance to rebel groups (Meier et al. (2023); Berlin and Malone (2023)), suggesting that the type of ES provided differs based on the recipients. Support dynamics further vary by conflict intensity, geographic proximity, and the relative strength of involved parties (Goldman and Abulof (2016); Meulewaeter (2021); Olson Lounsbery (2016); Salehyan et al. (2011)). The Cold War intensified conflicts through superpower support, but its end shifted patterns, reducing insurgencies and prolongation, while rebel groups lost aid, leading many conflicts to peter out or end in settlements (Testerman (2015); Kalyvas and Balcells (2010) in Roberts (2019)). While previous studies have established correlations between characteristics of conflicts and conflict parties, and quite a few have done so using data from the UCDP (Meulewaeter (2021); Salehyan et al. (2011); Olson Lounsbery (2016); Testerman (2015); Karlén (2017); Le et al. (2022)), most analysed the correlation between only one or two conflict characteristics and ES. Most studies further focus on the economic and social cost of conflict (e.g. Federle et al. (2024); Testerman (2015); Le et al. (2022); Sivard (1996)), or on the effects of ES on conflict dynamics, rather than the provision of ES itself. The introduction of the UCDP ESD (Meier et al. (2023)) offers an extensive descriptive overview of ES dynamics and an opportunity for extensive quantitative analysis of ES, making it easier to research ES dynamics. However, questions as to when, why, and how states are likely to intervene remain to be researched.

Previous research shows that external supporters engage in civil conflicts for diverse and strategic reasons, including geopolitical, economic, and ideological motives, as well as specific foreign policy objectives (Findley and Teo (2006); Meier et al. (2023)). Economic interests often drive involvement, with external powers seeking to protect trade benefits or access resources by backing domestic victors (Regan (1998); Kathman (2011)). In some instances, involvement appears benevolent, such as facilitating peace talks (Bhattarai (2016)). However, third-party intervention is rarely altruistic; states often aim to weaken adversaries, prevent conflict spillover, or maintain influence in post-colonial regions (Cunningham (2010); Gregory (2000)). Strategically, sponsoring militant groups enables states to destabilise rivals or gain leverage without direct conflict. Delegating violence minimises costs while exploiting the instability inflicted on adversaries (Schultz (2010); Bapat (2012)). However, this approach carries risks, including retaliation, escalation, and the possibility that armed groups will pursue independent agendas (Schultz (2010); Maoz and San-Akca (2012)). Shared ethnic or ideological ties between sponsors and armed groups signal shared goals and reduce the risk that resources will be misused (Berlin and Malone (2023); Salehyan (2010)). They are therefore among the strongest predictors of ES (Salehyan et al. (2011); Sozer (2016)), as they further facilitate communication and enhance trust, improving cooperation (San-Akca (2016); Bacon (2018)). However, relying on such ties can oversimplify sponsorship decisions, especially in contexts with multiple potential recipients or competing priorities (Berlin and Malone (2023)). Ultimately, prior research suggests that ES is a calculated foreign policy tool used to influence conflict outcomes while advancing sponsors' strategic goals.

Limitations of past research

The study of ES in ACs has provided critical insights but is hindered by several limitations. Much research focuses predominantly on support for non-state armed groups, particularly rebel organisations, sidelining

other actors and minor armed engagements (Walter (2004); Quinn et al. (2007); Salehyan et al. (2011)). A focus on rebel groups introduces methodological challenges and selection bias, limiting generalisability to other armed organisations (Abrahms et al. (2018); Phillips (2019)). This neglect obscures patterns of ES across diverse settings, undermining frameworks linking organisational attributes to state sponsorship. Furthermore, the multi-actor dimension of support is frequently overlooked, despite evidence that sponsors selectively support specific groups within fragmented conflicts (Berlin and Malone (2023)). Studies often emphasise the initial decision to delegate support (Salehyan (2010); San-Akca (2016)), neglecting how support dynamics evolve throughout a conflict's lifecycle and the strategic factors influencing its continuation or cessation (Karlén (2019); Tokdemir et al. (2021)). The content of interventions remains underexplored, despite its impact on conflict trajectories. Differentiating between support types - military training, material aid, or safe havens - can highlight distinct effects on outcomes (Roberts (2019)). Moreover, qualitative approaches dominate the literature, often at the expense of quantitative or mixed method analyses better suited to capturing temporal and contextual variations in support dynamics (Petrova (2019)). This research addresses these limitations by examining the interplay between conflict characteristics and different types of ES from a quantitative criminological angle, aiming to provide a starting point for further criminological discussions around the liability and responsibility of external supporters in ACs. By analysing trends in ES provision both within and between conflict-dyads, it further builds on previous research, offering a more nuanced understanding of the strategic factors influencing ES.

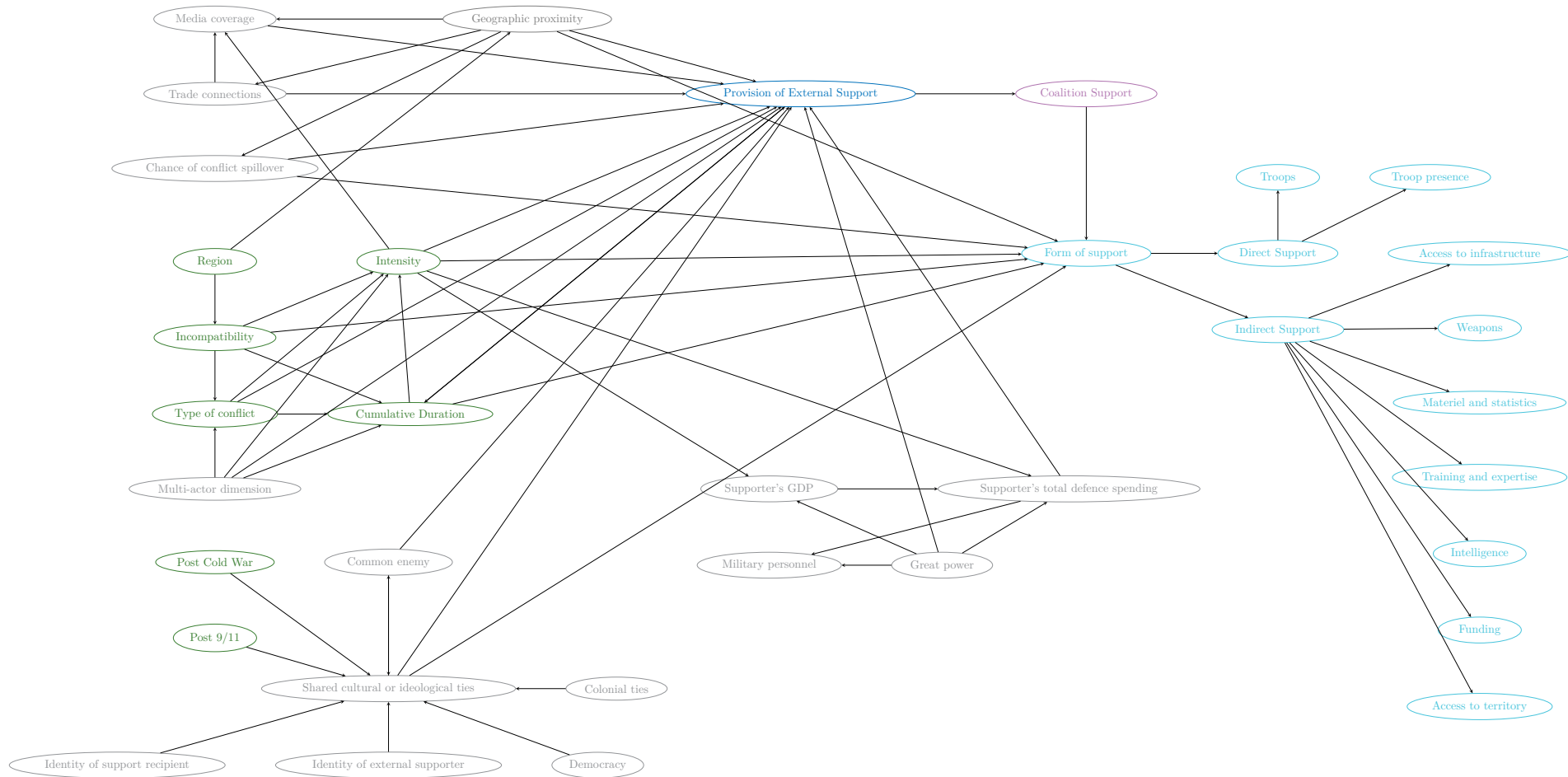


Figure 1: Directed Acyclic Graph on Conflict Characteristics and External Support.

OliveGreen = Independent variables (H1 and H2); Orchid = Independent variable (H3); RoyalBlue = Dependent variable (H1); SkyBlue = Dependent variables (H2 and H3); Gray = Not included in the analysis.

Based on the outlined Literature, Figure 1 portrays the underlying assumptions and relationships between conflict characteristics and ES to guide empirical analysis. Directed Acyclic Graphs serve as visual tools to encode theoretical assumptions about causal relationships, and guiding variable selection in statistical models. They map out the assumed pathways through which observed and unobserved factors influence the outcome, in this case, the provision and form of ES (Fleischer and Diez Roux (2008)). The dependent variable ‘provision of ES’ is assumed to be both directly and indirectly influenced by independent conflict characteristics (incompatibility, type of conflict, intensity, region, cumulative duration, point in time of the conflict), as well as the identity of the external supporter, the support recipient and their relationship with one another (i.e., colonial ties, shared cultural/ideological ties, a common enemy) and economic and political factors such as trade connections.

Research Design

The study aims to test several hypotheses regarding the relationship between conflict characteristics and ES. These hypotheses are grouped based on the type of relationship they examine, focusing on whether conflict characteristics influence the provision of ES, the form of support provided, and the impact of support coalitions on the type of assistance. These conflict characteristics are considered to be strategic factors influencing the provision of ES. Thereby, this study aims to examine whether external supporters follow a ‘cold’, rational approach to ES. The hypotheses are as follows:

Hypothesis Group 1 (HG1): Conflict Characteristics affect whether ES is provided to a conflict party.

- a) Conflict intensity affects whether ES is provided.
- b) Territorial incompatibility affects whether ES is provided.
- c) The cumulative duration of conflict affects whether ES is provided.
- d) The type of conflict affects whether ES is provided.
- e) Whether a conflict is taking place before or after 9/11 affects whether ES is provided.
- f) Whether a conflict is taking place during or after the Cold War affects whether ES is provided.

Hypothesis Group 2 (HG 2): Conflict characteristics influence what kind of ES is provided.

- a) Conflict intensity affects what kind of ES is provided.
- b) Territorial incompatibility affects what kind of ES is provided.
- c) The cumulative duration of conflict affects what kind of ES is provided.
- d) The type of conflict affects what kind of ES is provided.
- e) Whether a conflict is taking place before or after 9/11 affects what kind of ES is provided.
- f) Whether a conflict is taking place during or after the Cold War affects what kind of ES is provided.

Hypothesis 3 (H3): Whether support is provided in a coalition or as bilateral support affects what kind of ES is provided.

Based on Figure 1 HG1a–HG1f test whether OliveGreen conflict characteristics directly influence the RoyalBlue “Provision of External Support”. In HG2a–f, the same conflict characteristics are hypothesised to influence specific SkyBlue forms of support. H3 tests whether the Orchid “Coalition Support” node influences the type of ES (SkyBlue). It is important to note that Figure 1 is simply used for visual aid and

not as a strict representation of the model. The relationships between conflict characteristics and ES are not necessarily linear or direct, and the model does not imply causation. Instead, it serves as a framework for understanding the complex interplay between conflict characteristics and ES. While the model captures the relationships between conflict characteristics and ES, it does not account for all potential confounding factors, such as the identity of the external supporter, the support recipient, and their relationship with one another. These factors are represented in gray, indicating that they are not included in the analysis but may still influence the provision and form of ES. A further limitation is that it does not capture the temporal dynamics of conflicts, which could influence the provision and kind of ES. The model assumes that conflict characteristics have a direct impact on ES, but it does not account for potential feedback loops or interactions between variables. For example, the provision of ES could also influence conflict characteristics, such as conflict intensity, creating a cyclical relationship. Despite these limitations, the model provides a structured framework for testing the hypotheses outlined above.

Methodology

Data and operationalisation

This study draws on two key datasets from the Uppsala Conflict Data Program (UCDP): the UCDP Dyadic dataset (Version 18.1) (Harbom et al. (2008); Pettersson and Eck (2018)) and the UCDP External Support dataset (ESD) (Version 18.1) (Meier et al. (2023)). The UCDP Dyadic dataset provides data on ACs, focusing on dyads — pairs of primary warring parties — and spans from 1946 to 2018, with the unit of analysis being the dyad-year. The dataset includes 2,935 observations across 25 variables, capturing various conflict characteristics such as the type of conflict, incompatibilities, and combatants. The ESD, covering the period from 1975 to 2017, focuses specifically on ES provided to conflict parties, detailing ten types of support such as military assistance, training, and funding. It offers 2,272 observations of 96 variables, coded from open-source material and subjected to intercoder reliability checks.

The two datasets were merged using the common identifiers ‘dyad_id’ and ‘year’. Prior to merging, the datasets were checked for uniqueness to ensure the absence of duplicate rows. The merging process ensured the removal of duplicates, retaining only the necessary columns for analysis. The underlying unit of analysis remains the dyad-year. The final dataset includes 2234 observations of 123 variables on 472 dyads in 212 conflicts from 1975–2017. The comprehensive information on both conflict characteristics and characteristics of ES allows the study to address research questions about the relationship between the two and the types of support provided by external actors.

Despite their comprehensive nature, these datasets present several limitations. The availability of data on ES restricts the analysis to 1975–2017, excluding recent geopolitical developments. However, the period includes major historical phases - such as the Cold War, the post-Cold War era, and the aftermath of 9/11 - during which ES provision changed significantly (Testerman (2015); Kalyvas and Balcells (2010) in Roberts (2019); Meier et al. (2023)). The large sample enables comparative analysis across time, regions, and conflict types, which would be difficult to replicate through primary data collection. Still, UCDP’s state-based definition of AC includes only incompatibilities over government or territory, potentially overlooking underlying causes. Conflicts with fewer than 25 battle-related deaths per year are excluded, as are cases with uncertainty around key variables like incompatibility, actors, or intensity. Geographic precision is also lacking, as location is defined by the government side in a dyad (Themnér (2018)) rather than where the conflict occurs. The ESD only records first-degree supporters - those directly aiding a conflict party -, while second-degree supporters (e.g., states providing logistical assistance or funding regional coalitions) are omitted (Meier (2022)). Although the dataset distinguishes ten types of support, the categorisation simplifies complex, evolving interactions. Non-state troop support is consistently coded as zero, and support from international organisations, diaspora groups, businesses, or religious institutions is excluded (Meier (2022)). Lastly, the ESD’s minimum threshold for recording assistance may result in underreporting minor interventions, potentially affecting the accuracy of conclusions about ES provision.

The study incorporates dependent, independent, and control variables (Table 1), categorised based on their

measurement scales. All nominal independent variables were factorised to improve analytical clarity. Conflict characteristics were kept as per their original categorisation, while cumulative duration was derived using a time-based formula. The order of conflict type was changed so that 'intrastate' instead of 'interstate' would be the reference group to allow for clearer interpretation in conducted regression analyses. The dependent variables measuring ES, as defined in the Literature Review (see Meier et al. (2023); Meier (2022)), were expanded to differentiate between types of support (ext_category, ext_type).

Table 1: Variables used in the analysis

Category	Variable	Description	Measurement Scale	Proportion / Mean
Dependent Variables	ext_sup	External support provided (1 = Yes)	Binary	81.6
	ext_x	Troop support (1 = Yes)	Binary	18.93
	ext_p	Foreign troop presence (1 = Yes)	Binary	1.97
	ext_y	Access to infrastructure/joint operations (1 = Yes)	Binary	22.61
	ext_w	Weapons support (1 = Yes)	Binary	58.59
	ext_m	Materiel and logistics support (1 = Yes)	Binary	56.62
	ext_t	Training and expertise support (1 = Yes)	Binary	61.59
	ext_f	Funding support (1 = Yes)	Binary	46.42
	ext_i	Intelligence support (1 = Yes)	Binary	16.43
	ext_l	Access to territory (1 = Yes)	Binary	33.8
	ext_o	Other support (1 = Yes)	Binary	7.34
	ext_u	Unknown support (1 = Yes)	Binary	4.57
	ext_category	Categorisation of the form of external support provided (i.e., whether no support, one individual form or several forms of support are provided)	Ordinal	72.65
	ext_type	Categorisation of external support as direct, indirect, or no support	Nominal	
		Indirect		60.92
		Direct		0.98
		Direct and indirect		19.07
Independent Variables	type	Type of conflict (1 = Extrasystemic)	Nominal	
		Intrastate (3)		78.74
		Interstate (2)		3.13
		Internationalised intrastate (4)		18.13
	intensity	Conflict intensity (1 = Minor conflict, 2 = War)	Binary	20.1
	incompatibility	Conflict incompatibility	Nominal	
		Territory (1)		45.48
		Government (2)		54.03
		Territory and government (3)		0.49
	territorial	Conflict incompatibility is about territory (1 = Yes)	Binary	45.97
	cumulative_duration	Cumulative years of conflict (Years since first observed conflict year)	Ratio	7.892
	region	Region of conflict	Nominal	
		Europe (1)		5.33
		Middle East (2)		14.28
		Asia (3)		37.51
		Africa (4)		33.44
		Americas (5)		9.27
Control Variables				

Table 1: Variables used in the analysis (*continued*)

Category	Variable	Description	Measurement Scale	Proportion / Mean
	cold_war	Cold War status (0 = Cold War, 1 = Post-Cold War)	Binary	58.55
	nine_eleven	Period relative to 9/11 (0 = Before 9/11, 1 = After 9/11)	Binary	35
	ext_coalition	Coalition support (0 = Bilateral support, 1 = Coalition support)	Binary	6.22

Missing data analysis identified few cases of missing values for the variables used in the analysis. However, while the underlying data and computed variables cover most independent factors included in the theoretical framework outlined above (Figure 1), several important control variables are missing. These include geographic proximity, which influences cross-border spillover effects and logistical feasibility of support (Federle et al. (2024)), and major power status, as powerful states are more likely to intervene due to strategic interests (Goldman and Abulof (2016)). Other omitted factors include military capability differences, historical adversarial relationships, trade dependencies, former colonial ties, and the ideological alignment between external supporters and violent non-governmental organisations. The exclusion of these variables limits the study's ability to fully capture why certain conflicts receive ES while others do not.

Data analysis

Descriptive statistical methods were employed to summarise and analyse the data. This included the calculation of total conflict counts per year, as well as the summarisation of conflict intensity and the mean and median conflict durations by region. These statistics were visualised through various graphical methods, providing a clear depiction of temporal trends in conflict characteristics and support dynamics.

Regression analysis, utilising within-between (WB) models (Fairbrother (2014); Mundlak (1978)), was conducted to investigate factors influencing the provision of various forms of ES in AC. WB serves as a hybrid for random (RE) and fixed effects (FE). While FE control for conflict dyad-level heterogeneity by accounting for within-dyad variation, thus removing time-invariant confounders specific to each dyad (Huntington-Klein (2023)), RE assume individual effects are drawn from a random distribution (typically normal), which allows for estimating both within- and between-dyad variation, increasing precision and reducing standard errors (Huntington-Klein (2023)). The decision to use WB, capturing both dimensions, was driven by the need to account for both within-dyad and between-dyad variations, as this enables a more comprehensive assessment of how conflict characteristics are associated with external support, addressing both specific within-conflict dynamics and broader patterns across conflicts. For each form of support, WB logistic regression models were fitted to test the hypotheses that conflict characteristics and coalition support influence ES provision. As the dependent variables are binary, binomial logistic regression was considered the most appropriate approach. The core independent variables included incompatibility, conflict type, intensity, cumulative duration, coalition support, and historical periods such as the post-Cold War and post-9/11 eras. Region was initially included but later excluded due to high multicollinearity with conflict type (Appendix 3), which led to convergence issues in several models. Preliminary analyses and variable assessment (Table 1) supported the overall functional validity of the dependent variables. However, the 'incompatibility' variable was recoded to focus on territorial disputes, avoiding instability caused by low case counts in the combined 'territory and government' category.

UPDATE EQUATIONS All final models were estimated based on the corresponding *within or between* logistic regression equations:

WB Logistic Regression (based on RE equation):

$$\log \left(\frac{\text{provision of support}}{\text{no provision of support}} \right) = \alpha_i + \beta_1 \cdot \text{territorial incompatibility} + \beta_2 \cdot \text{type} + \beta_3 \cdot \text{intensity} + \beta_4 \cdot \text{duration} + \beta_5 \cdot \text{nine eleven} + \beta_6 \cdot \text{cold war} + \beta_7 \cdot \text{coalition support}$$

where $\log \left(\frac{\text{provision of support}}{\text{no provision of support}} \right)$ is the natural logarithm of the odds of the dependent variable occurring, α_i is the intercept varying between dyads (i), and β is the coefficient for each independent variable.

As models for the overall provision of support, troop support, foreign troop presence, and training and expertise failed to converge, they were re-estimated using Poisson Pseudo Maximum Likelihood (PPML) models. PPML is widely endorsed in the methodological literature for producing consistent estimates under conditions where logistic models with FE often encounter separation or quasi-complete separation issues (Silva and Tenreiro (2006)). It is also robust to heteroscedasticity and handles high-dimensional FE efficiently -

an important feature for longitudinal dyadic conflict data (Correia et al. (2020)). Given these advantages, PPML offers a methodologically sound alternative to logistic regression in contexts prone to convergence problems. The refitted PPML models used the same set of independent variables as the logistic regressions, with the dependent variable defined as the provision of each respective form of external support and were estimated using the following equation:

WB PPML Regression:

$$\mathbb{E}(Y_i | X_i, \alpha_i) = \exp \left(\alpha_i + \beta_1 \cdot \text{Territorial Incompatibility}_i + \beta_2 \cdot \text{Type}_i + \right. \\ \left. \beta_3 \cdot \text{Intensity}_i + \beta_4 \cdot \text{Duration}_i + \beta_5 \cdot \text{Post 9/11}_i + \right. \\ \left. \beta_6 \cdot \text{Cold War}_i + \beta_7 \cdot \text{Coalition Support}_i \right)$$

where $\mathbb{E}(Y_{it} | X_{it}, \alpha_i)$ is the expected count outcome (e.g. probability of receiving support) for unit i , conditional on covariates and random effect, and α_i is the dyad-specific random effect. These analyses provide an overview of trends in the provision of ES without making inferential claims about causal relationships.

Model fit was assessed throughout using the Bayesian Information Criterion (BIC), Akaike Information Criterion (AIC), with lower values indicating better fit (Gelman and Hill (2006)). An overall comparison of conflict characteristics' effects on ES was conducted between and within conflicts, with troop support and weapons models selected for detailed analysis. These were chosen due to the strength of their associations with conflict characteristics. They also enabled exploration of how different conflict characteristics correlate with the provision of direct versus indirect support. However, the use of distinct model types across support forms (due to convergence issues) limits the direct comparability between direct and indirect support provisions.

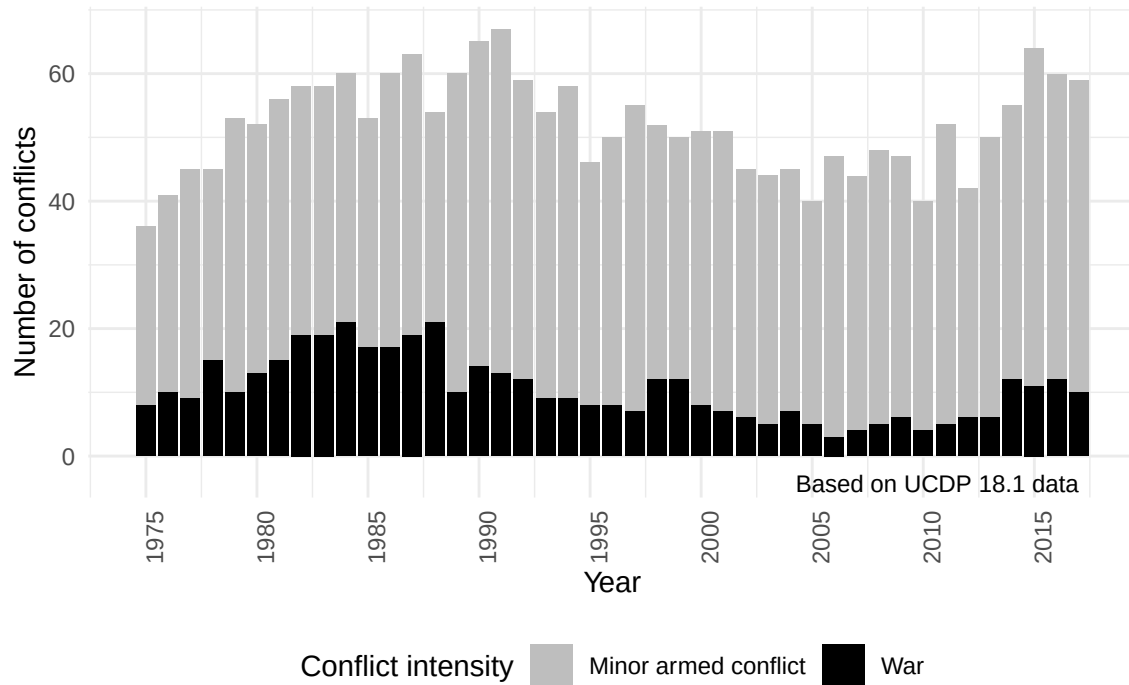
Logistic regression results were interpreted using odds ratios, reflecting the change in odds of receiving support for a one-unit increase in the predictor. PPML results were interpreted through incidence rate ratios (IRR), indicating the change in expected count of ES provision for a one-unit increase in the independent variable. While the dependent variables are binary, PPML models treat them as count variables—an accepted compromise to allow valid estimation. To enhance interpretability, logistic regression coefficients were exponentiated. Despite discussions on redefining statistical significance (Benjamin et al. (2018)), this report rejects the null hypothesis that the population's estimated coefficient is zero when $p < .05$. Multicollinearity was not expected due to the distinct nature of the conflict characteristics. Nevertheless, variance inflation factors (VIFs) were calculated. Standard regression assumptions were tested across models to ensure validity and reliability. However, as most predictors were categorical, the linearity assumption did not apply to them, and due to the logistic nature of the models, homoscedasticity and residual normality were not relevant.

Findings

Overall trends in conflict characteristics

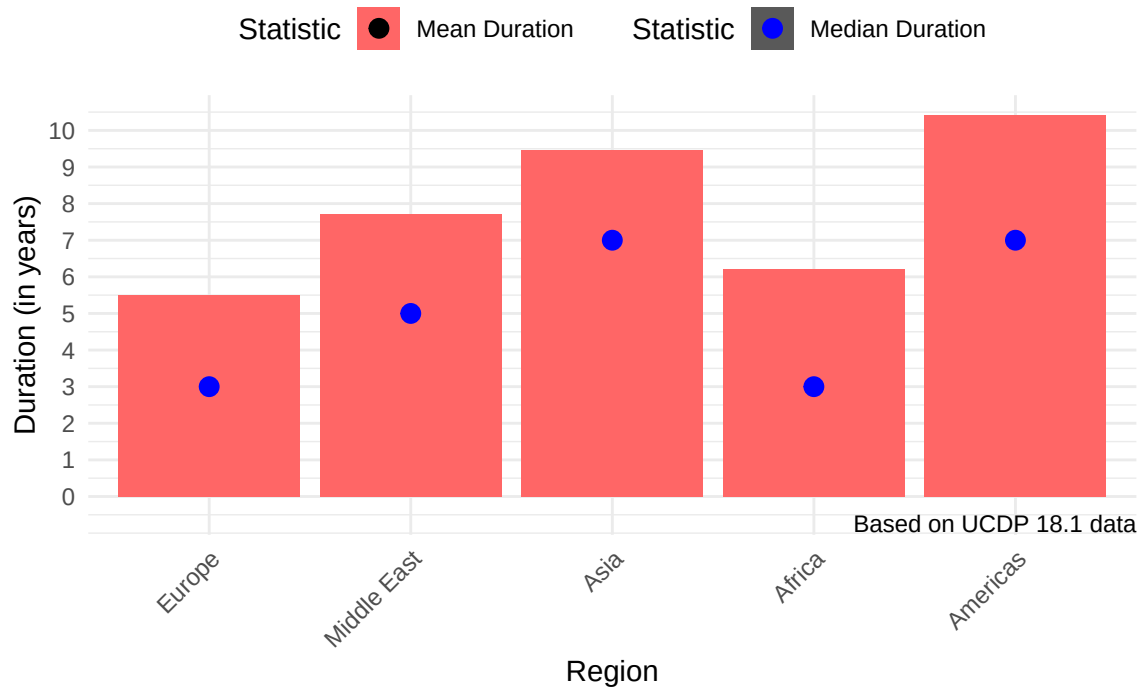
Although the total number of conflicts has decreased and minor ACs make up the majority of ACs, recent years have seen a rise in wars (Figure 2).

Figure 2: World armed conflicts by intensity level



While most conflicts are fought in Asia and Africa, the number of conflicts fought in the Middle East is growing (Appendix 1), with conflicts in Asia and the Americas portraying longer duration (Figure 3).

Figure 3: Mean and Median Conflict Duration by Region



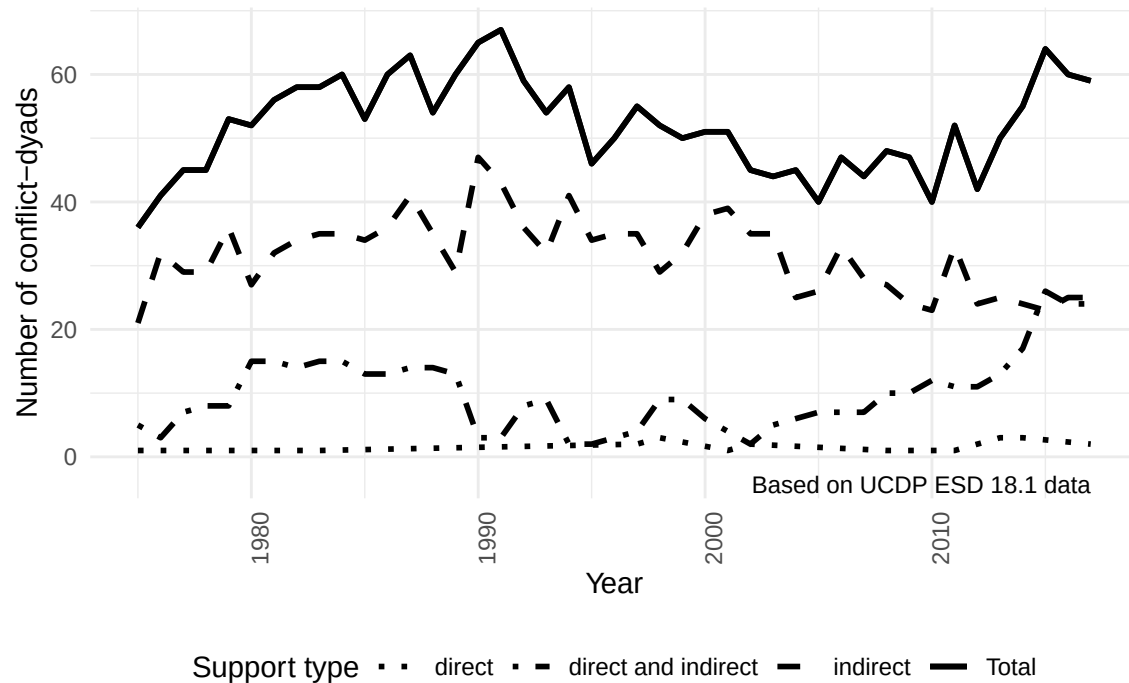
Intrastate conflicts are the most common type of conflict, followed by internationalised intrastate conflicts, whose number has been growing since the early 2000s (Appendix 2). While there is fluctuation between the

incompatibility types ‘territory’ and ‘government’, most conflicts in 2017 were about territory and hardly any about territory and government (Appendix 4).

The provision of external support

The provision of ES has risen, with indirect support being the most common (Figure 4) and supporters providing three forms of support on average (Median: 4). The provision of support in a coalition has increased over time (Appendix 5).

Figure 4: Direct and Indirect Support, 1975–2017



The model for the provision of ES (Table 2) shows no statistically significant results. However, the results of the regression analyses for individual forms of support reveal distinct patterns in the relationship between conflict characteristics and the provision of ES, with notable differences between within and between models.

Table 2: Regression Results – External Support Determinants

	Within-between
(Intercept)	1.004 [0.862, 1.170]
Type (interstate)	1.004 [0.540, 1.867]
Type (internationalised intrastate)	1.005 [0.865, 1.166]
AIC	2707.6
BIC	2785.6
min_wave	0
max_wave	42
N	208
model	within-between

Table 3a and 3b reveal important differences in how conflict characteristics correlate with the provision of various forms of ES, both within conflicts (i.e., changes over time within the same dyad) and between conflicts (i.e., differences across dyads). The provision of ES is less correlated to conflict characteristics within conflicts (Table 3a) than between conflicts (Table 3b) and the effects of conflict characteristics on the provision of ES tend to be smaller, with all effect sizes below 15 within conflicts.

The era post-9/11, as well as conflict intensity and duration, show the most significance across within models. However, the effect of these characteristics on the provision of ES within conflicts is very small, with only the effect of the 9/11 on the provision of weapons (8.06) and the effect of conflict intensity on materiel and statistics provision (11.11) exceeding an effect size of 5. Conflict intensity and duration are only correlated with the provision of indirect ES but not with the provision of direct support.

Within conflict-dyads, the provision of weapons, funding, and intelligence are correlated with most conflict characteristics, with conflict intensity, duration, and the post-9/11 era being significant in all three. The provision of troop support is not correlated to any conflict characteristics, while troop presence is only correlated to conflict duration. Training and expertise is not correlated to any conflict characteristics and the provision of materiel and statistics, access to infrastructure, and access to territory are only correlated to one or two conflict characteristics. All conflict characteristics only show small effects (below 15 in all and below 5 in all but three cases) on ES provision within conflict-dyads.

In the between logistic regression (Table 3b), the type of conflict — particularly internationalised intrastate conflicts — and conflict intensity emerge as strongly correlated to the provision of ES, showing significance in six (intensity) and seven (type) of the nine models. While the effect of conflict intensity on the provision of different forms of ES is small to moderate (reaching from an effect of 0.15 on access to territory to an effect of 19.09 on the provision of funding), the effect of conflict type on ES provision ranges from small to high, depending on the form of support. I.e., while the odds of receiving training and expertise are only 0.29 units higher compared to intrastate conflicts, the odds of receiving troop support are 87.72 units higher compared to intrastate conflicts, holding all other factors constant. Other influential factors include conflict duration, and the post-9/11 period, which are strongly correlated with multiple forms of support. While the effect of conflict duration stays small (below 2 in all cases) between conflict-dyads, 9/11 portrays an extremely high effect of 191.83 on intelligence provision and a high effect on access to infrastructure (17.97) and troop support (19.69) between conflict-dyads.

Foreign troop presence, weapons, and materiel and statistics provision appear to be the most sensitive to conflict characteristics, with three to four out of seven independent variables reaching statistical significance. Troop support, access to infrastructure, training and expertise, funding, and access to territory all show correlation to two conflict characteristics, while intelligence is only correlated to one conflict characteristic. While not influenced by many conflict characteristics the effects of conflict characteristics on troop support and intelligence are the highest among all models. While troop support is correlated to the type of conflict, troop presence is further correlated to conflict intensity, duration and the post-9/11 period.

Table 3a: Statistical Significance in the Within Logistic and PPML Regression Models

Independent Variable	Troop Support	Troop Presence	Access to Infrastructure	Weapons	Materiel and Statistics	Training and Expertise	Funding	Intelligence	Access to Territory
Intercept									
Territorial dispute									
Type of conflict									
Intensity				** (War)	* (War)		*** (War)	** (War)	
Duration		*		**			***	**	
9/11			** (After)	*** (After)			* (After)	* (After)	* (After)
Cold War				* (After)				* (After)	
Coalition Support			* (Coalition)		. (Coalition)		* (Coalition)		
N	208	208	208	208	208	208	208	208	208
AIC	877.15	282.08	1109.38	935.08	1046.81	2641.1	1144.73	823.69	1168.21
BIC	955.15	360.08	1187.37	1013.08	1124.8	2719.09	1222.72	901.69	1246.21

Table 3b: Statistical Significance in the Between Logistic and PPML Regression Models

Independent Variable	Troop Support	Troop Presence	Access to Infrastructure	Weapons	Materiel and Statistics	Training and Expertise	Funding	Intelligence	Access to Territory
Intercept	***	***	***	***		***		***	*
Territorial dispute									
Type of conflict	*** (Interstate); *** (Internationalised intrastate)	. (Internationalised intrastate)	** (Internationalised intrastate)	. (Internationalised intrastate)	*** (Internationalised intrastate)	* (Interstate)	* (Internationalised intrastate)		
Intensity		** (War)		** (War)	** (War)	* (War)	*** (War)		. (War)
Duration		*			*				*
9/11		* (After)	** (After)					*** (After)	
Cold War				*** (After)					
Coalition Support									
N	208	208	208	208	208	208	208	208	208
AIC	877.15	282.08	1109.38	935.08	1046.81	2641.1	1144.73	823.69	1168.21
BIC	955.15	360.08	1187.37	1013.08	1124.8	2719.09	1222.72	901.69	1246.21

Already presented in Table 3, the models for troop support and weapons were selected for detailed analysis to explore how different conflict characteristics correlate with the provision of direct versus indirect support. While both models use a within-between specification, the model for troop support employs a PPML regression, and the model for weapons provision uses a logistic regression.

The provision of troop support is most strongly correlated with the type of conflict in the between model, while the provision of weapons shows broader but more varied correlations with conflict characteristics in both model components (Table 4). Specifically, between conflict dyads, the IRR of receiving foreign troop support is 25.8 units higher ($p < 0.001$) for interstate conflicts and 87.72 units higher ($p < 0.001$) for internationalised intrastate conflicts, compared to intrastate conflicts, holding all other variables constant. In contrast, the between model for weapons provision finds that conflict type is less consistently significant, with only internationalised intrastate conflicts nearing significance and showing moderately higher odds (odds = 2.31, $p = 0.05$) for weapon support, while interstate conflicts show no significant association.

The intensity of a conflict is significantly correlated with weapons provision in both within and between models. In the between model, wars are associated with 13.75 higher odds ($p = 0.01$) of receiving weapons than minor conflicts, and in the within model, the odds increase by 2.88 units ($p = 0.01$) when a conflict escalates to war, controlling for all dyad-level variation. In contrast, conflict intensity is not significantly correlated with troop support in either component of the model. The duration of conflict also exhibits different patterns across models. For weapons, the within model shows that the odds of receiving weapons increase by 0.92 units ($p = 0.01$) with each additional year in a conflict. No such within-group effect emerges in the troop support model. In the between model, neither outcome is significantly associated with the average duration of conflict.

Temporal conflict markers also yield divergent results. Both models indicate that conflicts occurring after the Cold War are correlated with the provision of weapons, but not with troop support. Controlling for all between-dyad variation, the odds of receiving weapons increase by 0.2 units ($p = 0.02$) after the Cold War compared to during the war within each dyad. In the between component of the model, conflicts with a higher proportion of years occurring during the Cold War period were associated with significantly lower odds of receiving weapons support (odds = 0.04, $p < 0.001$). The within model for weapons provision indicates that conflicts occurring after 9/11 are associated with higher odds (odds = 8.06, $p < 0.001$) of receiving weapons than conflicts happening before 9/11, while this effect is absent in the between model. No comparable effect is found in the troop support model.

Neither model finds statistically significant effects for coalition-based support or territorial incompatibility in either the within or between components. In sum, while troop support appears to be driven largely by structural characteristics of the conflict, particularly its type, the provision of weapons is more responsive to dynamic factors within conflicts, including duration, intensity, and geopolitical shifts such as the post-9/11 context.

Table 4: WBM Logistic Regression Results

	Troop Support	Weapons
(Intercept)	0.012*** [0.006, 0.025]	32.049*** [6.970, 147.360]
Type (interstate)	25.803*** [6.793, 98.016]	0.116 [0.000, 36.836]
Type (internationalised intrastate)	87.715*** [45.671, 168.466]	2.310* [1.018, 5.241]
AIC	877.2	935.1
BIC	955.2	1013.1
min_wave	0	0
max_wave	42	42
N	208	208
model	within-between	within-between
pR2_fe		0.20243076168103
pR2_total		0.809335039727066

All models were tested against standard regression assumptions to ensure validity and reliability. UCDP data was deemed appropriate for addressing the research questions, as it comprehensively captures AC and ES characteristics, although empirical datasets rarely meet all theoretical criteria perfectly (Gelman and Hill (2006)). Table 1 confirmed the functionality of the dependent variables. Multicollinearity was assessed using VIFs, with all values below 5, indicating no significant collinearity issues (O'Brien (2007)). As most predictors were categorical, the linearity assumption did not apply. The assumption of independent errors was violated due to repeated observations per conflict and dyad. Homoscedasticity and residual normality were not applicable to logistic models. The binary nature of the dependent variable confirmed the appropriateness of logistic regression for the analysis.

Discussion

The findings of this study align with existing literature on conflict dynamics and external interventions. Over time, the number of wars has slightly increased, indicating a shift towards more intense conflicts, while conflict duration has increased, suggesting that many conflicts persist rather than being resolved. These trends are consistent with research highlighting the protracted nature of modern ACs (Quinn et al. (2007)). The nature of ES has also evolved. While troop support has declined, the total number of external interventions continues to rise (Figure 4), demonstrating a persistent strategic interest in shaping conflict trajectories through both direct and indirect means (Davies et al. (2024); Chang and Sellak (2022)). Recent years have seen a shift toward multilateral coalitions and collaborative interventions (Appendix 5), with nearly one-third of all support instances involving such coalitions, reflecting a greater emphasis on burden-sharing and legitimacy (Meier et al. (2023)).

The findings from the regression analysis suggest that while structural conflict factors are correlated to the provision of ES, temporal and contextual elements play a decisive role within individual conflicts. This aligns with existing literature on the provision of ES, which emphasises the importance of conflict characteristics in shaping external intervention strategies (Berlin and Malone (2023); Davies et al. (2024)). While the models examining HG1 show no statistically significant results, regression analyses for individual forms of support reveal distinct patterns in the relationship between conflict characteristics and the provision of ES, with notable differences in the within and between results. The magnitude of effect sizes across models contributes an additional layer to these findings, complementing statistical significance and offering insights into the substantive impact of various predictors.

The relatively small effects of conflict characteristics on ES provision within conflict-dyads suggest that although factors such as duration, conflict intensity, and the post-9/11 period are statistically significant in some models, their practical influence on the provision of ES within dyads may be modest. Exceptions include the post-9/11 period's effect on weapons provision, and conflict intensity's effect on the provision of materiel and statistics, indicating a heightened responsiveness of certain forms of indirect support to geopolitical shifts. The larger and more variable effect sizes, especially for the type of conflict and temporal markers between conflict-dyads suggest that conflict type is a key driver of direct military engagement across conflict contexts.

While troop support displays high odds ratios in the between model, it is only significantly associated with conflict type, suggesting a narrow but powerful predictor. Conversely, access to infrastructure, weapons, and materiel and statistics — forms of indirect support — show broader sensitivity to multiple conflict characteristics and are often accompanied by moderate effect sizes. This pattern reinforces findings from existing scholarship which argues that indirect ES is increasingly favoured in modern interventions due to its strategic flexibility and lower political cost (Meier et al. (2023); Berlin and Malone (2023)). While no trends could be discovered in the provision of direct forms of support and the correlation to different conflict characteristics of troop presence and troop support might seem counter intuitive at first, it could indicate an escalation in the provision of ES with the former being a precondition for the latter. Conflict intensity, the post-9/11 period, conflict type (between) and conflict duration (within and between) might be first incentives for external supporters to provide more direct support but once troops are already present, the type of conflict might then become a more important factor in the decision to provide troop support.

Therefore, HG1 cannot be completely rejected but the results rather indicate that the provision of ES is not only correlated to conflict characteristics but that it may further be dependent on the form of support provided.

For HG2, which explores whether conflict characteristics influence the type of ES provided, results reveal that conflict intensity (H2a) is significantly correlated to the provision of weapons, materiel and statistics, and funding in both within and between models, as well as to troop presence, training and expertise, intelligence, and access to territory in either within or between conflict-dyads, albeit with varying effect sizes. This aligns with the argument that higher-intensity conflicts are more likely to receive ES due to concerns about spillover effects and escalation (Cunningham (2010); Gregory (2000)). However, intensity does not appear to significantly impact the provision of troop support, suggesting that external actors may not relate the intensity of a conflict to increased levels of direct support.

Besides intensity, conflict duration (H2c) appears as strongly correlated to the provision of weapons, funding, intelligence, and troop presence within a conflict-dyad, and to troop presence, the provision of materiel and statistics and access to territory between conflict-dyads. While correlated to the provision of several forms of ES, the effect of duration remains small in all models. The period after 9/11 (H2e) is correlated to the provision of multiple forms of indirect support — particularly access to infrastructure, weapons, funding, intelligence, and access to territory — within conflicts, and to troop presence, access to infrastructure and intelligence in the between results. The striking between-model effects of the post-9/11 period's influence on intelligence provision and access to infrastructure underline the substantial shift in intervention strategies after this geopolitical turning point. These high-magnitude effects illustrate that temporal context can sometimes outweigh even structural characteristics, especially in shaping forms of support associated with intelligence, and logistical facilitation. The Cold War period (H2f) is correlated to the provision of weapons and intelligence within conflicts and to weapons across conflicts. These findings are consistent with research highlighting the geopolitical significance of the Cold War and post-9/11 eras, during which intervention strategies shifted from ideological proxy wars to more targeted and strategic forms of engagement (Testerman (2015); Kalyvas and Balcells (2010) in Roberts (2019)). The type of conflict (H2d) — particularly internationalised intrastate and interstate conflicts — is significantly correlated with ES in the between models, specifically troop support, troop presence, access to infrastructure, weapons, materiel and statistics, training and expertise, and funding. On the other hand, territorial incompatibility (H2b) is not significantly correlated with the provision of any ES, suggesting that this factor plays a less prominent role in shaping external intervention strategies. The lack of significant correlations between territorial incompatibility with ES provision is surprising, as previous research has highlighted the importance of this factor in shaping conflict dynamics and external interventions (Cunningham (2010); Gregory (2000)).

Findings suggest that coalition support (H3) is significantly correlated to access to infrastructure, materiel and statistics, and funding within conflict-dyads. However, between conflict-dyads, it shows no statistical significance at all. Coalition support further only yields modest effects, suggesting that coalitional dynamics influence tactical decisions within specific conflicts more than strategic choices across conflicts.

Model-specific results further illustrate these patterns: troop support is most strongly correlated to the type of conflict, whereas the provision of weapons shows broader but less consistent correlations with conflict characteristics. The results suggest that the type of conflict could be a significant factor in determining the provision of troop support between conflicts, with interstate and internationalised intrastate conflicts being more likely to receive this form of support than intrastate conflicts. The significance of especially internationalised intrastate conflicts might be explained by the fact that they are often more complex and involve multiple actors, which creates opportunities for external actors to provide support to one side or another, depending on their interests and objectives. The findings highlight key theoretical implications for Criminology and conflict studies. The weak correlation between direct support and conflict characteristics suggests that direct military interventions may be driven by factors beyond conflict characteristics alone, such as diplomatic or geopolitical considerations. While statistical significance flags robust relationships, the size of these effects clarifies their substantive impact. The findings suggest that some conflict characteristics exert strong influences on particular forms of support — notably conflict type on troop support and post-9/11 on intelligence — while others like territorial incompatibility show little predictive power despite theoretical expectations. For researchers and policymakers, this highlights the importance of not only identifying which

variables matter but also understanding how much they matter, ensuring that theoretical claims are grounded in both statistical and substantive relevance. The significance of temporal markers and conflict characteristics across both dimensions suggests that the dynamics shaping ES are multifaceted and vary depending on the form of assistance.

The inability to control for regional variation or geographic proximity limits the generalisability of the findings, as well as their interpretation from a criminological perspective. Tobler (1970) law, ‘everything is related to everything else, but near things are more related than distant things’ suggests that geographic proximity may play a significant role in shaping the dynamics of ES. Not controlling for geographic proximity might thereby ignore important contextual factors that influence the provision of ES. Despite this, the results confirm existing theories that ES provision is influenced by conflict characteristics — particularly intensity and duration — yet also underscore that different forms of support respond to these characteristics in distinct ways. This supports a ‘cold logic’ interpretation, in which state actors rationally assess the strategic value of conflict before engaging, rather than reacting emotionally. This aligns with prior research suggesting intervention decisions are based on a rational assessment of costs, benefits, and political alignment (Schultz (2010); Bapat (2012); Sawyer et al. (2015)). However, the absence of data on shared cultural ties or geographic proximity prevents assessment of whether certain interventions also follow a ‘hot’ logic, where emotional or identity-driven factors — such as shared ethnicity or history — play a role. This opens an important avenue for future research that more explicitly incorporates emotional, symbolic, and identity-based motivations.

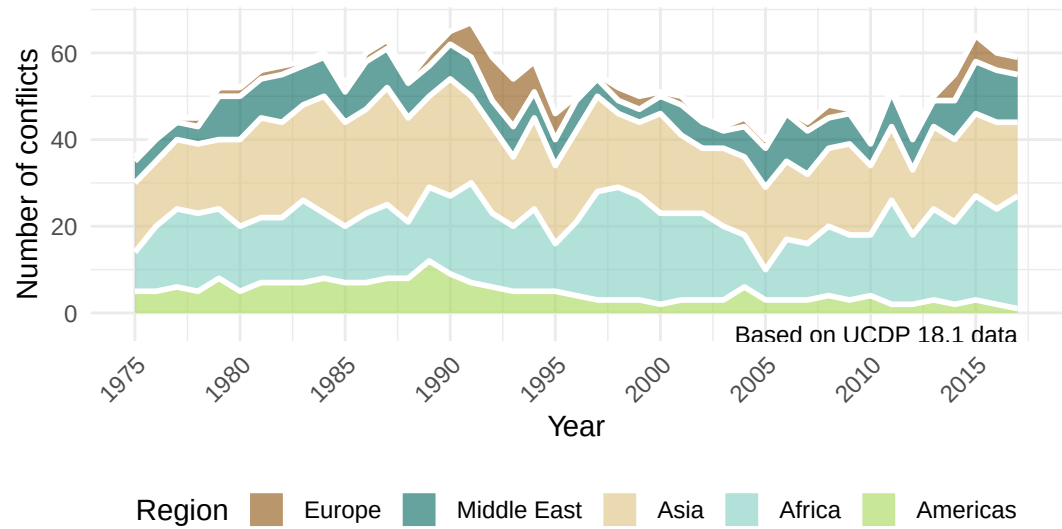
Further limitations arise from the dataset’s focus on state-based conflicts, which omits the growing influence of non-state actors and transnational networks in modern conflict. Additionally, provider and recipient characteristics — key to understanding the criminological dynamics of power, ideology, and legitimacy — could not be tested due to data constraints. Reverse causality also poses a risk: rather than resolving conflict, ES may prolong it by altering the incentives for continued violence (Fearon (2004) in Testerman (2015); Lacina (2006)). While model refitting and variable exclusion (i.e., 9/11 in the FE model for foreign troop presence) were necessary to achieve convergence, these changes limit comparability across models. The exclusive use of quantitative methods further restricts insight into the lived experiences and micro-level effects of ES. Future research should therefore enhance this study by incorporating qualitative data and case studies to provide a more comprehensive understanding of the relationship between ES and conflict dynamics.

Conclusion

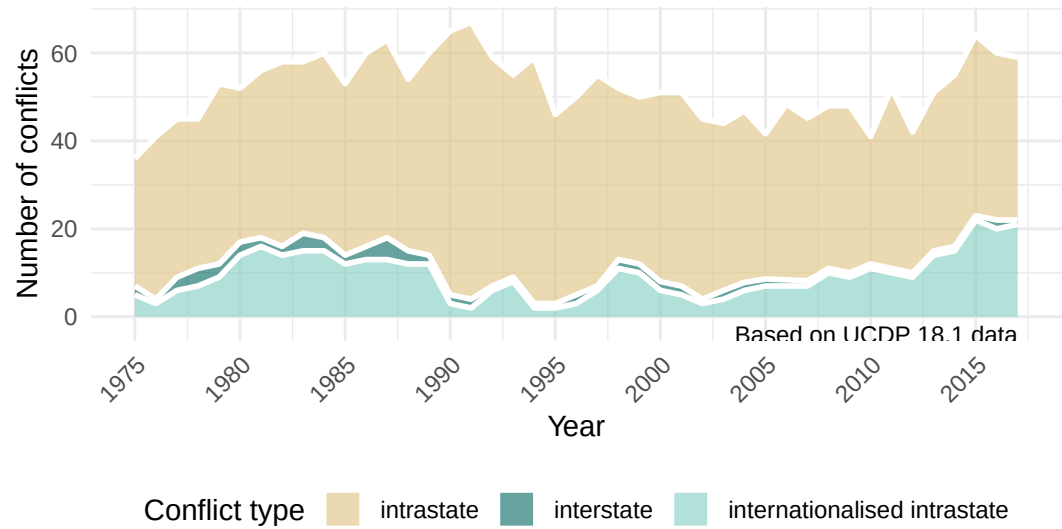
This study set out to examine how characteristics of AC shape the provision of ES, offering new insights into the strategic logic underlying state interventions. Contemporary examples such as Russia’s invasion of Ukraine, evidence the limitations of pacifist responses. But while military interventions can at times stabilise volatile regions, they often risk escalating violence or prolonging conflict (Fearon (2004) in Testerman (2015); Lacina (2006); Olson Lounsbery (2016)). The findings show that ES—particularly indirect forms—is significantly associated with conflict characteristics, potentially suggesting a calculated, ‘cold’ approach to intervention. Rather than being random or purely humanitarian, ES would then reflect strategic assessments of conflict dynamics, reinforcing existing literature on the instrumental use of support in warfare (Schultz (2010); Bapat (2012); Sawyer et al. (2015)). Crucially, the type of support provided varies with conflict characteristics, indicating that ES is not a one-size-fits-all strategy. These results have important implications for criminology. If states make informed, strategic decisions to intervene in armed conflicts, they may bear greater responsibility for the violence and crimes that unfold in these contexts. Yet, this dimension of liability and complicity remains underexplored in criminological research. This study offers a starting point for engaging with ES as a site of criminological concern, where questions of accountability, harm, and power intersect. By broadening its focus to include the indirect consequences of state interventions, criminology can contribute to more effective policy responses and a deeper understanding of the global production and regulation of violence.

Appendices

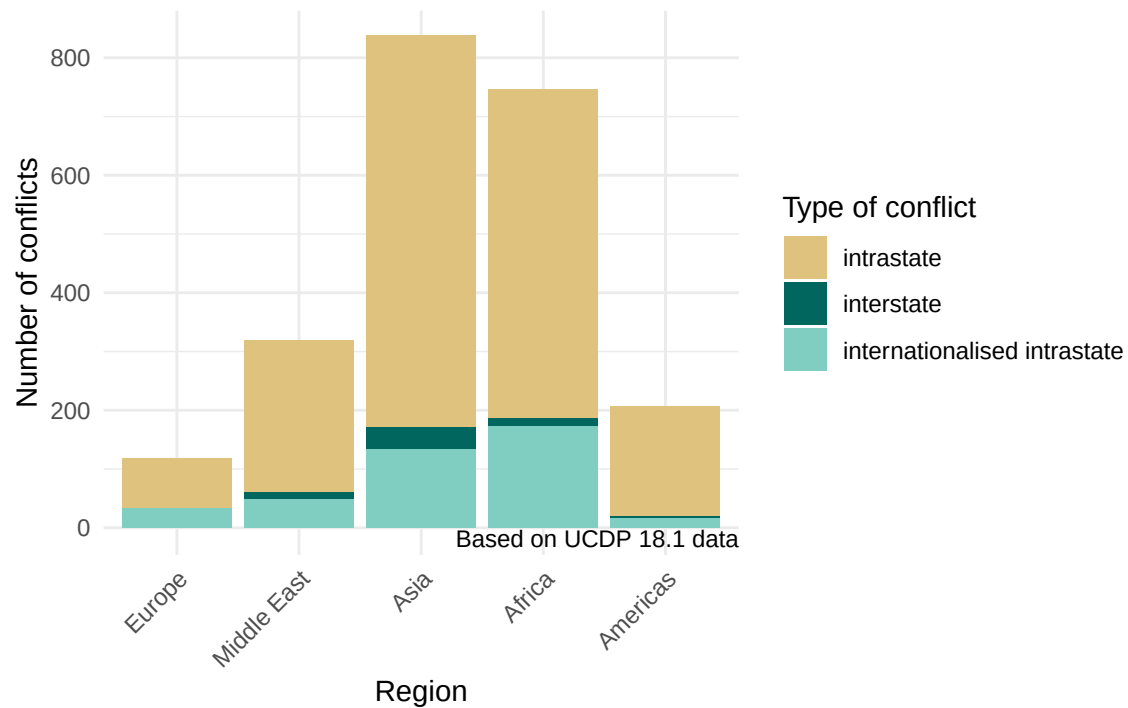
Appendix 1: State-based Conflicts by Region, 1975–2017



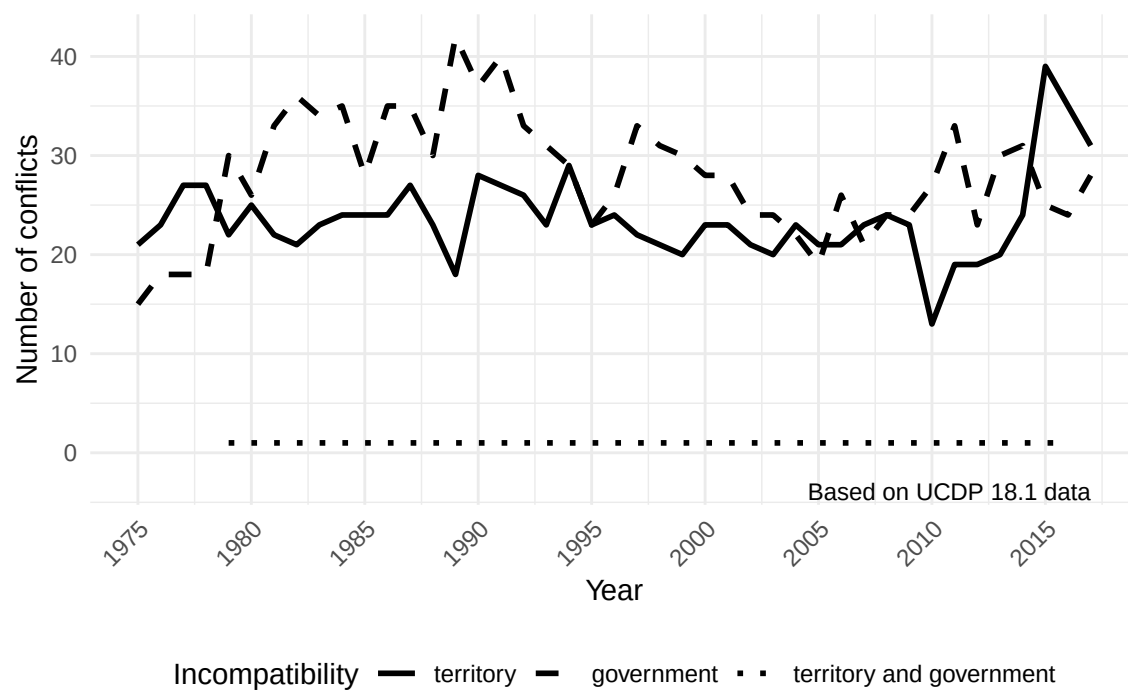
Appendix 2: State-based Conflicts by Type of Conflict, 1975–2017



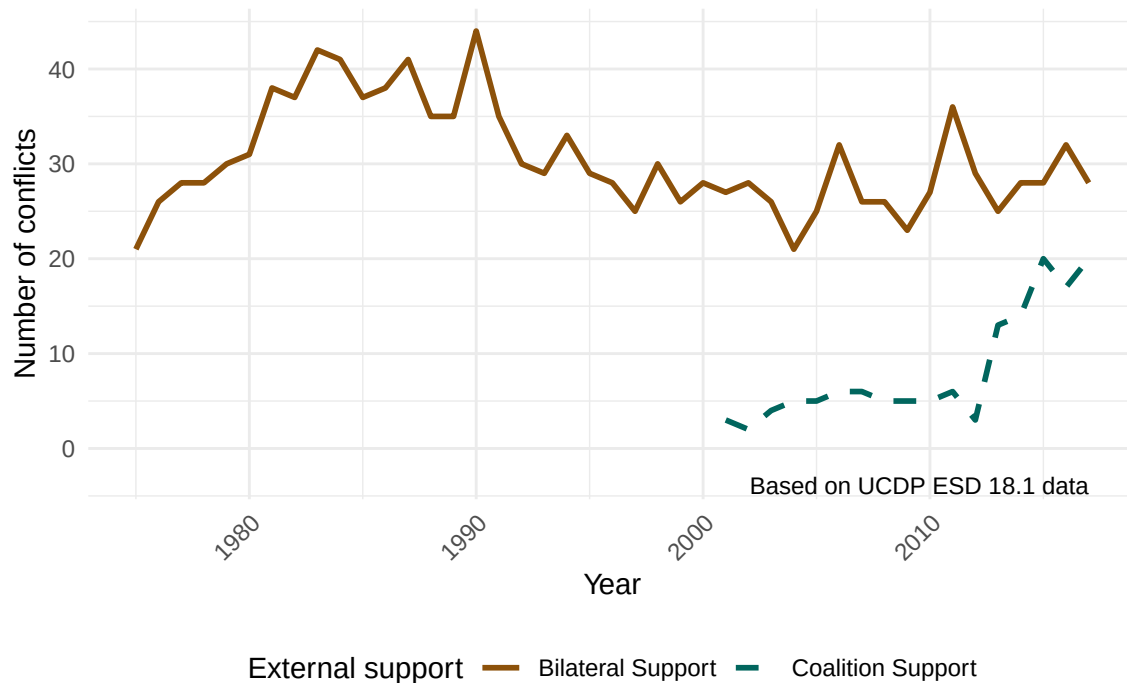
Appendix 3: Conflict Type by Region



Appendix 4: Evolution of State-based Conflicts by Incompatibility, 1975–2017



Appendix 5: Evolution of External Support: Bilateral vs Coalition



References

- Abrahms, M., Ward, M. and Kennedy, R. (2018). Explaining civilian attacks: Terrorist networks, principal-agent problems and target selection. *Perspectives on terrorism (Lowell)*, 12(1), pp.23–45. [online]. Available from: <https://www.jstor.org/stable/26343744> [Accessed April 21, 2025].
- Anderson, N. (2019). Competitive intervention, protracted conflict, and the global prevalence of civil war. *International studies quarterly*, 63(3), pp.692–706.
- Aydin, A. and Regan, P.M. (2012). Networks of third-party interveners and civil war duration. *European journal of international relations*, 18(3), pp.573–597.
- Bacon, T. (2018). Is the enemy of my enemy my friend?: How terrorist groups select partners. *Security studies*, 27(3), pp.345–378.
- Balch-Lindsay, D., Enterline, A.J. and Joyce, K.A. (2008). Third-party intervention and the civil war process. *Journal of peace research*, 45(3), pp.345–363.
- Bapat, N.A. (2012). Understanding state sponsorship of militant groups. *British journal of political science*, 42(1), pp.1–29.
- Benjamin, D.J. et al. (2018). Redefine statistical significance. *Nature human behaviour*, 2(1), pp.6–10.
- Berlin, M. and Malone, I. (2023). Go arm me: How militant fragmentation affects external support. *International interactions*, 49(4), pp.557–586.
- Bhattarai, P. (2016). Third-party coordination in conflict resolution: Evidence from nepal and the philip-

pines. *The International journal of conflict management*, 27(3), pp.398–423.

Chang, Y.-M. and Sellak, M. (2022). A theory of competing interventions by external powers in intrastate conflicts: Implications for war and armed peace. *Applied economics*, 54(33), pp.3811–3822.

Collins, R. (2008). *Violence : A micro-sociological theory / randall collins*. Princeton, N.J. ; Princeton University Press.

Correia, S., Guimarães, P. and Zylkin, T. (2020). Fast poisson estimation with high-dimensional fixed effects. *The Stata Journal*, 20(1), pp.95–115. [online]. Available from: <https://doi.org/10.1177/1536867X20909691>.

Cunningham, D.E. (2010). Blocking resolution: How external states can prolong civil wars. *Journal of peace research*, 47(2), pp.115–127.

Cunningham, D.E. (2016). PREVENTING CIVIL WAR: How the potential for international intervention can deter conflict onset. *World politics*, 68(2), pp.307–340.

Davies, S. et al. (2024). Organized violence 1989–2023, and the prevalence of organized crime groups. *Journal of peace research*, 61(4), pp.673–693.

Fairbrother, M. (2014). Two multilevel modeling techniques for analyzing comparative longitudinal survey datasets. *Political science research and methods*, 2(1), pp.119–140.

Fang, X. et al. (2020). The economic consequences of conflict in sub-saharan africa. *IMF working paper*, 20(221).

Federle, J. et al. (2024). *The price of war*. Centre for Economic Policy Research. [online]. Available from: https://mutschler.eu/files/papers/Price_of_War_2024_CEPR.pdf [Accessed April 21, 2025].

Findley, M.G. and Teo, T.K. (2006). Rethinking third-party interventions into civil wars: An actor-centric approach. *The Journal of politics*, 68(4), pp.828–837.

Fleischer, N.L. and Diez Roux, A.V. (2008). Using directed acyclic graphs to guide analyses of neighbourhood health effects: An introduction. *Journal of epidemiology and community health (1979)*, 62(9), pp.842–846.

Gelman, A. and Hill, J. (2006). *Data analysis using regression and multilevel/hierarchical models*. Cambridge University Press.

Gleditsch, K.S. (2007). Transnational dimensions of civil war. *Journal of peace research*, 44(3), pp.293–309.

Goldman, O.S. and Abulof, U. (2016). Democracy for the rescue-of dictators? The role of regime type in civil war interventions. *Contemporary security policy*, 37(3), pp.341–368.

Gregory, S. (2000). THE FRENCH MILITARY IN AFRICA: PAST AND PRESENT. *African affairs (London)*, 99(396), pp.435–448.

Hagan, J. (2015). While criminology slept: A criminal war of aggression in iraq. *The Criminologist: The Official Newsletter of the American Society of Criminology*, 40(6), pp.2–4. [online]. Available from: <https://asc41.org/wp-content/uploads/ASC-Criminologist-2015-11.pdf> [Accessed April 21, 2025].

Harbom, L., Melander, E. and Wallensteen, P. (2008). Dyadic dimensions of armed conflict, 1946–2007. *Journal of peace research*, 45(5), pp.697–710.

Hillyard, P. and Tombs, S. (2007). From ‘crime’ to social harm? *Crime, Law and Social Change*, 48, pp.9–25.

- Hoeffler, A. and Reynal-Querol, M. (2003). *Measuring the costs of conflict*. Centre for the Study of African Economies, University of Oxford; The World Bank. [online]. Available from: https://v1.cepa.lk/content_images/publications/documents/1005-S-Hoeffler%20and%20Reynal-Measuring%20the%20Costs%20of%20Conflict.pdf [Accessed April 21, 2025].
- Huntington-Klein, N. (2023). Fixed effects. In *The effect: An introduction to research design and causality*. CRC Press, A Chapman & Hall Book. [online]. Available from: <https://theeffectbook.net/ch-FixedEffects.html>.
- Jamieson, R. (2003). Introduction. *Theoretical Criminology*, 7(3), pp.259–263. [online]. Available from: <https://doi.org/10.1177/13624806030073001>.
- Karlén, N. (2023). Escalate to de-escalate? External state support and governments' willingness to negotiate. *Studies in conflict and terrorism*, 46(8), pp.1323–1344.
- Karlén, N. (2017). The legacy of foreign patrons: External state support and conflict recurrence. *Journal of peace research*, 54(4), pp.499–512.
- Karlén, N. (2019). Turning off the taps: The termination of state sponsorship. *Terrorism and political violence*, 31(4), pp.733–758.
- Kathman, J.D. (2011). Civil war diffusion and regional motivations for intervention. *Journal of Conflict Resolution*, 55(6), pp.847–876.
- Klein, J.R. (2011). Toward a cultural criminology of war. *Social justice (San Francisco, Calif.)*, 38(3), pp.86–103. [online]. Available from: <https://www.jstor.org/stable/41940949> [Accessed February 13, 2025].
- Kramer, R.C. and Michalowski, R.J. (2005). WAR, AGGRESSION AND STATE CRIME: A criminological analysis of the invasion and occupation of Iraq. *British journal of criminology*, 45(4), pp.446–469.
- Lacina, B. (2006). Explaining the severity of civil wars. *The Journal of conflict resolution*, 50(2), pp.276–289.
- Le, T.-H., Bui, M.-T. and Uddin, G.S. (2022). Economic and social impacts of conflict: A cross-country analysis. *Economic modelling*, 115, pp.105980–.
- Maoz, Z. and San-Akca, B. (2012). Rivalry and state support of non-state armed groups (NAGs), 1946-2001 1: Rivalry and state support of non-state armed groups. *International studies quarterly*, 56(4), pp.720–734.
- McGarry, R. et al. (2016). Introduction: The criminology of war, what is it good for? In *The palgrave handbook of criminology and war*. United Kingdom: Palgrave Macmillan UK, pp. 1–21.
- McGarry, R. and Walklate, S. (2019). Introduction: Can there be a 'criminology of war'? In *A criminology of war?* New horizons in criminology. Bristol, UK: Bristol University Press, pp. 1–14.
- McGarry, R. and Walklate, S. (2015). Introduction: Placing war within criminology. In S. Walklate & R. McGarry, eds. *Criminology and war: Transgressing the borders*. Abingdon: Routledge. [online]. Available from: <https://www.taylorfrancis.com/chapters/edit/10.4324/9781315858425-1/introduction-ross-mcgarry-sandra-walklate?context=ubx&refId=6b6a1a3e-4b01-4767-8838-da05005a9a2e> [Accessed April 21, 2025].
- Meier, V. et al. (2023). External support in armed conflicts: Introducing the UCDP external support dataset (ESD), 1975–2017. *Journal of peace research*, 60(3), pp.545–554.
- Meier, V. (2022). UCDP external support dataset (ESD) codebook v18.1.

- Meulewaeter, C. (2021). Military expenditure, arms transfer and armed conflicts. In J. C. Rufanges, ed. *Military spending and global security*. United Kingdom: Routledge, pp. 22–40.
- Mundlak, Y. (1978). On the pooling of time series and cross section data. *Econometrica*, 46(1), pp.69–85.
- O'Brien, R.M. (2007). A caution regarding rules of thumb for variance inflation factors. *Quality & quantity*, 41(5), pp.673–690.
- Olson Lounsbury, M. (2016). Foreign military intervention, power dynamics, and rebel group cohesion. *Journal of global security studies*, 1(2), pp.127–141.
- Petrova, M.G. (2019). What matters is who supports you: Diaspora and foreign states as external supporters and militants' adoption of nonviolence. *The Journal of conflict resolution*, 63(9), pp.2155–2179.
- Pettersson, T. and Eck, K. (2018). Organized violence, 1989–2017. *Journal of peace research*, 55(4), pp.535–547.
- Phillips, B.J. (2019). Terrorist group rivalries and alliances: Testing competing explanations. *Studies in conflict and terrorism*, 42(11), pp.997–1019.
- Quinn, J.M., Mason, T.D. and Gurses, M. (2007). Sustaining the peace: Determinants of civil war recurrence. *International interactions*, 33(2), pp.167–193.
- R Core Team. (2025). R: The r project for statistical computing.
- Rafter, N.H. (2016). *The crime of all crimes: Toward a criminology of genocide*. New York: New York University Press.
- Rauchhaus, R.W. (2009). Principal-agent problems in humanitarian intervention: Moral hazards, adverse selection, and the commitment dilemma. *International studies quarterly*, 53(4), pp.871–884.
- Regan, P.M. (1998). Choosing to intervene: Outside interventions in internal conflicts. *The Journal of politics*, 60(3), pp.754–779.
- Regan, P.M. (2002). Third-party interventions and the duration of intrastate conflicts. *Journal of Conflict Resolution*, 46(1), pp.55–73.
- Regan, P.M. and Meachum, M.S. (2014). Data on interventions during periods of political instability. *Journal of peace research*, 51(1), pp.127–135.
- Roberts, J. (2019). Targeting and resistance: Reassessing the effect of external support on the duration and outcome of armed conflict. *Civil wars*, 21(3), pp.362–384.
- Ruggiero, V. (2005). Criminalizing war: Criminology as ceasefire. *Social & legal studies*, 14(2), pp.239–257.
- Ruggiero, V. (2023). Criminology against war. *Justice, Power and Resistance*, 6(3), pp.263–277.
- Ruggiero, V. (2015). War and the death of achilles. In S. Walklate & R. McGarry, eds. *Criminology and war*. Routledge, pp. 21–37.
- Saideman, S.M. (2002). Discrimination in international relations: Analyzing external support for ethnic groups. *Journal of peace research*, 39(1), pp.27–50.
- Salehyan, I. (2010). The delegation of war to rebel organizations. *The Journal of conflict resolution*, 54(3),

pp.493–515.

Salehyan, I., Gleditsch, K.S. and Cunningham, D.E. (2011). Explaining external support for insurgent groups. *International organization*, 65(4), pp.709–744.

San-Akca, B. (2016). *States in disguise : Causes of state support for rebel groups / belgin san-akca*. Oxford: Oxford University Press.

Sandefur, J. and Kenny, C. (2025). USAID cuts: New estimates at the country level. [online]. Available from: <https://www.cgdev.org/blog/usa-id-cuts-new-estimates-country-level> [Accessed May 7, 2025].

Sawyer, K., Cunningham, K.G. and Reed, W. (2015). The role of external support in civil war termination. *The Journal of conflict resolution*, 61(6), pp.1174–1202.

Schultz, K.A. (2010). The enforcement problem in coercive bargaining: Interstate conflict over rebel support in civil wars. *International organization*, 64(2), pp.281–312.

Silva, J.M.C.S. and Tenreyro, S. (2006). The log of gravity. *The review of economics and statistics*, 88(4), pp.641–658. [online]. Available from: <https://www.jstor.org/stable/40043025>.

Sivard, R.L. (1996). *World military and social expenditures*. 16th ed. Washington, D.C., USA: World Priorities. [online]. Available from: https://www.dropbox.com/s/wm76wylpwo5d851/W MSE%201996%20ed.%20%2816th%29%20by%20Ruth%20Sivard%20_300dpi.pdf?dl=0 [Accessed April 21, 2025].

Sozer, B. (2016). Development of proxy relationships: A case study of the lebanese civil war. *Small wars & insurgencies*, 27(4), pp.636–658.

Sykes, G.M. and Matza, D. (1957). Techniques of neutralization: A theory of delinquency. *American sociological review*, 22(6), pp.664–670.

Testerman, M. (2015). Removing the crutch: External support and the dynamics of armed conflict. *Studies in conflict and terrorism*, 38(7), pp.529–542.

Themnér, L. (2018). UCDP dyadic dataset codebook, version 18.1.

Tobler, W.R. (1970). A computer movie simulating urban growth in the detroit region. *Economic geography*, 46, pp.234–240.

Tokdemir, E. et al. (2021). Rebel rivalry and the strategic nature of rebel group ideology and demands. *The Journal of conflict resolution*, 65(4), pp.729–758.

Toukan, M. (2019). International politics by other means: External sources of civil war. *Journal of Peace Research*, 56(6), pp.812–826. [online]. Available from: <https://www.jstor.org/stable/48596232> [Accessed April 21, 2025].

UN. (2025). US aid cuts will make world ‘less healthy, less safe and less prosperous’: gutierrez. [online]. Available from: <https://news.un.org/en/story/2025/02/1160646> [Accessed May 7, 2025].

Walklate, S. and McGarry, R. (2019). Theorising ‘war’ within sociology and criminology. In *A criminology of war?* Policy Press, pp. 15–40.

Walter, B.F. (2004). Does conflict beget conflict? Explaining recurring civil war. *Journal of peace research*, 41(3), pp.371–388.